

# ICSC Ambassador Problem Set Solutions

## Problem 1: Hashing Basics

A hash function is used to convert data (such as text or numbers) into a fixed-size value called a hash code. In computer science, hash functions are commonly used in hash tables to quickly store and retrieve data. A good hash function is important because it distributes data evenly across the hash table. This reduces collisions, which happen when multiple inputs produce the same hash value. Fewer collisions make searching, inserting, and deleting data much faster and more efficient.

## Problem 2: Boolean Logic

The light turns on only when exactly one switch is on. Boolean expression:  $(A \text{ AND NOT } B \text{ AND NOT } C) \text{ OR } (\text{NOT } A \text{ AND } B \text{ AND NOT } C) \text{ OR } (\text{NOT } A \text{ AND NOT } B \text{ AND } C)$  This expression checks all cases where only one switch is turned on.

## Problem 3: Count Even Numbers

Algorithm: 1. Start with a counter set to 0. 2. Go through each number in the list. 3. If the number is divisible by 2, increase the counter by 1. 4. After checking all numbers, return the counter. Pseudo Code: `count = 0` for each number in list: `if number mod 2 == 0: count = count + 1` return count